implosion, there was no more mystery. An imploding star really does shrink through the critical circumference without hesitation. That it appears to freeze as seen from far away is an illusion.

The embedding diagrams of the parable of the ants capture only some of the insight that came from Finkelstein's new reference frame, not all. Further insight is embodied in Figure 6.7, which is a *spacetime diagram* for the imploding star.

Until now, the only spacetime diagrams we have met were in the flat spacetime of special relativity; for example, Figure 1.3. In Figure 1.3, we drew our diagrams from two different viewpoints: that of an inertial reference frame at rest in the city of Pasadena (with the downward pull of gravity ignored), Figure 1.3c; and that of an inertial frame attached to your high-speed sports car as you zoom down Pasadena's, Colorado Boulevard, Figure 1.3b. In each diagram we plotted our chosen frame's space horizontally, and its time vertically.

In Figure 6.7, the chosen reference frame is that of Finkelstein. Accordingly, we plot horizontally two of the three dimensions of space, as measured in Finkelstein's frame ("Finkelstein's space"), and we plot vertically time as measured in his frame ("Finkelstein's time"). Since, far from the star, Finkelstein's frame is static (not imploding), Finkelstein's time there is that experienced by a static observer. And since, near the star, Finkelstein's frame falls inward with the imploding stellar surface, Finkelstein's time there is that experienced by an infalling observer.

Two horizontal slices are shown in the diagram. Each depicts two of the dimensions of space at a specific moment of time, but with the space's curvature removed so the space looks flat. More specifically, circumferences around the star's center are faithfully represented on these horizontal slices, but radii (distances from the center) are not. To represent both radii and circumferences faithfully, we would have to use embedding diagrams like those of Figure 6.2 or those of the parable of the ants, Figure 6.6. The space curvature would then show clearly: Circumferences would be less than 2π times radii. By drawing the horizontal slices flat, we are artificially removing their curvature. This incorrect flattening of the space is a price we pay to make the diagram legible. The payoff we gain is our ability to see space and time together on a single, legible diagram.